OLIST E-COMMERCE

Dataset Overview & Project Description

Project: Olist E-Commerce Sales & Customer Experience Analytics

Team: InsightHub

Primary Tool: Microsoft Power BI

Document purpose
This document introduces the Olist dataset, summarizes the available tables, and defines the proposed analytical project scope. It is an initial planning document; findings and KPI values will be confirmed after data preparation and validation.

1. Project Description

Olist E-Commerce Sales & Customer Experience Analytics is a data analytics project based on the Brazilian E-Commerce Public Dataset by Olist. The project will examine sales performance, customer purchasing patterns, delivery operations, product categories, payment behavior, customer reviews, and seller performance.

The team will use Microsoft Power BI to prepare and model the data, build interactive reports, and communicate business insights. Power Query will support data cleaning and transformation, while DAX will support measures and calculations where appropriate.

Problem Statement

E-commerce operations generate data across orders, customers, products, payments, deliveries, reviews, and sellers. Analyzing these records separately can make it difficult to form a coherent view of business performance. This project will integrate relevant tables and examine patterns in sales, delivery outcomes, and customer feedback while accounting for data quality and table relationships.

Project Objectives

- Analyze order volume and sales value over time.
- Examine customer purchasing patterns and order value.
- Evaluate delivery times against estimated delivery dates.
- Analyze review-score distributions and customer feedback patterns.
- Compare performance across product categories.
- Explore seller performance and geographic distribution.
- Analyze payment methods, payment values, and installment patterns.
- Build interactive Power BI dashboards and provide evidence-based recommendations.

Proposed Scope

In scope: data inspection, cleaning and transformation, data modeling, descriptive analysis, KPI development, interactive Power BI reporting, interpretation, and recommendations.

Not assumed at this stage: predictive modeling, causal conclusions, or business results that have not been calculated and validated. Any extension beyond descriptive analysis will be decided after reviewing the data and project requirements.

2. Dataset Overview

The Brazilian E-Commerce Public Dataset by Olist is organized into related tables rather than one flat spreadsheet. Each table represents a different entity or level of detail. Row counts must therefore not be interpreted as the number of orders or customers without considering the table's grain.

Available Tables

Table	Purpose / Main Information	Approx. rows*
Customers	Customer identifiers and location: city, state, ZIP prefix.	99,441
Geolocation	ZIP prefixes with latitude, longitude, city, and state.	1,000,163
Order Items	Products in orders, seller IDs, item prices, freight values, shipping-limit dates.	112,650
Order Payments	Payment method, sequence, installments, and payment value.	103,886
Order Reviews	Review scores, optional comments, and review timestamps.	99,224
Orders	Order status and purchase, approval, carrier, delivery, and estimated-delivery dates.	99,441
Products	Product categories and physical attributes such as weight and dimensions.	32,951
Product Category Translation	Mapping from original category names to English names.	71
Sellers	Seller identifiers and seller location: city, state, ZIP prefix.	3,095

*Row counts are based on the initial file review and should be rechecked against the exact files used in the final model. Counts are not directly comparable because the tables have different levels of detail.

Key Relationships

- **Customers to Orders:** `customer_id`.
- **Orders to Order Items:** `order_id`.
- **Orders to Order Payments:** `order_id`.
- **Orders to Order Reviews:** `order_id`; confirm relationship cardinality.
- **Order Items to Products:** `product_id`.
- **Order Items to Sellers:** `seller_id`.
- **Products to Category Translation:** category name; review unmatched values.

Geolocation contains multiple records for some ZIP prefixes. It should not be joined as if each ZIP prefix were unique; consider a suitable lookup or aggregation if geographic analysis requires it.

3. Data Quality Considerations

The initial review identified areas that need validation during data preparation. These are not automatic reasons to delete records; handling should depend on the analysis objective and the meaning of each field.

Area	Consideration	Planned handling
Missing dates	Some orders lack approval, carrier, or customer-delivery timestamps.	Inspect by order status; do not invent missing dates.
Review comments	Review title/message may be blank even when a score exists.	Retain valid scores; treat text analysis separately.
Category mapping	Some product categories may not match the translation table.	Identify unmatched values and preserve original categories.
Repeated order IDs	Items and payments may contain multiple rows per order.	Use appropriate aggregation and avoid duplicated totals.
Geolocation grain	ZIP prefixes may have multiple coordinate records.	Create a suitable lookup/aggregation if needed.
Time coverage	Reviewed purchase dates span approximately Sep 2016 to Oct 2018, with uneven coverage at the boundaries.	Check monthly counts before interpreting trends or forecasting.

Important KPI and Modeling Notes

- Distinguish order count, item count, payment-record count, and unique-customer count.
- Do not treat payment values and item prices as the same measure.
- Define the sales/revenue metric explicitly before building KPI cards.
- Define delivery delay using actual and estimated timestamps, including a rule for canceled or undelivered orders.
- Validate relationships and test measures to detect accidental duplication.
- Use distinct customer identifiers when the objective is to count unique customers.

4. Proposed Business Questions

- How do order volume and sales value change over time?
- Which product categories contribute most to item sales value?
- What is the distribution of order statuses?
- How do actual delivery outcomes compare with estimated dates?
- What share of eligible delivered orders arrived after the estimated date?
- How are review scores distributed, and how do they vary by delivery outcome?
- Which payment methods are used, and how do payment values vary?
- How does seller performance vary by sales value and location?

Expected Deliverables

- Cleaned and transformed data prepared in Power Query.
- A documented Power BI data model with validated relationships.
- DAX measures and KPIs with clearly stated definitions.
- Interactive Power BI report pages for sales, customer experience, delivery, product, seller, and payment analysis, as supported by validated data.
- A summary of findings and practical business recommendations.
- Project documentation and final presentation.

5. Tools and Working Approach

Microsoft Power BI: data modeling, measures, reporting, and interactive visualization.
Power Query: importing, cleaning, shaping, and transforming source tables.
DAX: calculations and KPIs in the Power BI model.
GitHub: project documentation, README, and versioned project materials.

The proposed workflow is to inspect the source tables, define key metrics, clean and transform the data, build and test the data model, develop measures, create dashboards, and then formulate recommendations from verified results.

6. Project Status and Limitations

Status: In Progress — planning and data understanding. The scope and analysis areas are preliminary. Final KPIs, relationships, and report pages will be confirmed after data profiling and review of the official project requirements. No analytical results or business conclusions are asserted in this document.

Dataset reference: Brazilian E-Commerce Public Dataset by Olist. Update this document if the dataset version or project scope changes.